



# MATH130: Linear Equations and Functions

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## Learning Objectives

- Evaluate a linear function at a given input
- Compute the slope of a line and write its equation
- Solve linear inequalities and express the solution set
- Identify the domain and range of a function

*Simplify each expression completely. Show all steps and circle your final answer.*

## Evaluating a linear function

1. Given  $f(x) = 2x + -8$ , find  $f(-1)$ .

$$f(x) = 2x - 8, \quad x = -1$$

Answer: \_\_\_\_\_

2. Given  $g(x) = 2x - 5$ , find  $g(-2)$ .

$$f(x) = 2x - 5, \quad x = -2$$

Answer: \_\_\_\_\_

3. Given  $f(x) = 6x + -6$ , find  $f(1)$ .

$$f(x) = 6x - 6, \quad x = 1$$

Answer: \_\_\_\_\_

4. Given  $g(x) = 6x - 6$ , find  $g(-2)$ .

$$f(x) = 6x - 6, \quad x = -2$$

Answer: \_\_\_\_\_

5. Given  $f(x) = 5x + -7$ , find  $f(3)$ .

$$f(x) = 5x - 7, \quad x = 3$$

Answer: \_\_\_\_\_

6. Given  $g(x) = 6x - 4$ , find  $g(3)$ .

$$f(x) = 6x - 4, \quad x = 3$$

Answer: \_\_\_\_\_

7. Given  $f(x) = 1x + 3$ , find  $f(4)$ .

$$f(x) = 1x + 3, \quad x = 4$$

Answer: \_\_\_\_\_

8. Given  $g(x) = 6x - 4$ , find  $g(-1)$ .

$$f(x) = 6x - 4, \quad x = -1$$

Answer: \_\_\_\_\_

9. Given  $f(x) = 3x + -6$ , find  $f(-2)$ .

$$f(x) = 3x - 6, \quad x = -2$$

Answer: \_\_\_\_\_

10. Given  $g(x) = 5x - 8$ , find  $g(0)$ .

$$f(x) = 5x - 8, \quad x = 0$$

Answer: \_\_\_\_\_

11. Given  $f(x) = 2x + -5$ , find  $f(1)$ .

$$f(x) = 2x - 5, \quad x = 1$$

Answer: \_\_\_\_\_

12. Given  $g(x) = 7x - 8$ , find  $g(4)$ .

$$f(x) = 7x - 8, \quad x = 4$$

Answer: \_\_\_\_\_

### Solving linear inequalities

13. Solve the inequality:  $4x + 7 < 8$ . Express the answer as  $x < k$ .

$$4x + 7 < 8$$

Answer: \_\_\_\_\_

14. Solve the inequality:  $3x - 7 \geq 1$ . Express the answer as  $x \geq k$ .

$$3x - 7 \geq 1$$

Answer: \_\_\_\_\_

15. Solve the inequality:  $1x + 2 < 4$ . Express the answer as  $x < k$ .

$$1x + 2 < 4$$

Answer: \_\_\_\_\_

16. Solve the inequality:  $4x - 5 \geq -1$ . Express the answer as  $x \geq k$ .

$$4x - 5 \geq -1$$

Answer: \_\_\_\_\_

17. Solve the inequality:  $2x + 5 < 8$ . Express the answer as  $x < k$ .

$$2x + 5 < 8$$

Answer: \_\_\_\_\_

18. Solve the inequality:  $1x - 4 \geq -4$ . Express the answer as  $x \geq k$ .

$$1x - 4 \geq -4$$

Answer: \_\_\_\_\_

19. Solve the inequality:  $2x + -7 < 5$ . Express the answer as  $x < k$ .

$$2x + -7 < 5$$

Answer: \_\_\_\_\_

20. Solve the inequality:  $3x - 5 \geq 3$ . Express the answer as  $x \geq k$ .

$$3x - 5 \geq 3$$

Answer: \_\_\_\_\_

21. Solve the inequality:  $4x + -6 < 3$ . Express the answer as  $x < k$ .

$$4x + -6 < 3$$

Answer: \_\_\_\_\_

22. Solve the inequality:  $4x - 5 \geq -2$ . Express the answer as  $x \geq k$ .

$$4x - 5 \geq -2$$

Answer: \_\_\_\_\_

23. Solve the inequality:  $6x + -8 < -6$ . Express the answer as  $x < k$ .

$$6x + -8 < -6$$

Answer: \_\_\_\_\_

24. Solve the inequality:  $1x - 5 \geq 5$ . Express the answer as  $x \geq k$ .

$$1x - 5 \geq 5$$

Answer: \_\_\_\_\_

### Slope between two points

**25.** Find the slope of the line passing through  $(-4, 4)$  and  $(-3, 3)$ .

$(-4, 4), (-3, 3)$

Answer: \_\_\_\_\_

**26.** Find the slope of the line passing through  $(-2, 8)$  and  $(1, 7)$ .

$(-2, 8), (1, 7)$

Answer: \_\_\_\_\_

**27.** Find the slope of the line passing through  $(3, -8)$  and  $(11, 0)$ .

$(3, -8), (11, 0)$

Answer: \_\_\_\_\_

**28.** Find the slope of the line passing through  $(2, -3)$  and  $(6, -1)$ .

$(2, -3), (6, -1)$

Answer: \_\_\_\_\_

**29.** Find the slope of the line passing through  $(-4, 1)$  and  $(3, 0)$ .

$(-4, 1), (3, 0)$

Answer: \_\_\_\_\_

**30.** Find the slope of the line passing through  $(0, 8)$  and  $(1, 8)$ .

$(0, 8), (1, 8)$

Answer: \_\_\_\_\_



# MATH130: Linear Equations and Functions

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ANSWER KEY & SOLUTIONS

*Topics: Solving linear inequalities, Slope between two points, Evaluating a linear function. All answers verified by independent computation.*

## Solutions

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## Evaluating a linear function

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1. Given  $f(x) = 2x + -8$ , find  $f(-1)$ .

$$f(x) = 2x - 8, \quad x = -1$$

→ Substitute  $x = -1$ :  $f(-1) = 2(-1) + -8 = -2 + -8 = -10$ .

**Answer:**  $f(-1) = -10$

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2. Given  $g(x) = 2x - 5$ , find  $g(-2)$ .

$$f(x) = 2x - 5, \quad x = -2$$

→ Substitute  $x = -2$ :  $g(-2) = 2(-2) - 5 = -4 - 5 = -9$ .

**Answer:**  $f(-2) = -9$

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3. Given  $f(x) = 6x + -6$ , find  $f(1)$ .

$$f(x) = 6x - 6, \quad x = 1$$

→ Substitute  $x = 1$ :  $f(1) = 6(1) + -6 = 6 + -6 = 0$ .

**Answer:**  $f(1) = 0$

---

4. Given  $g(x) = 6x - 6$ , find  $g(-2)$ .

$$f(x) = 6x - 6, \quad x = -2$$

→ Substitute  $x = -2$ :  $g(-2) = 6(-2) - 6 = -12 - 6 = -18$ .

**Answer:**  $f(-2) = -18$

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5. Given  $f(x) = 5x + -7$ , find  $f(3)$ .

$$f(x) = 5x - 7, \quad x = 3$$

→ Substitute  $x = 3$ :  $f(3) = 5(3) + -7 = 15 + -7 = 8$ .

**Answer:**  $f(3) = 8$

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6. Given  $g(x) = 6x - 4$ , find  $g(3)$ .

$$f(x) = 6x - 4, \quad x = 3$$

→ Substitute  $x = 3$ :  $g(3) = 6(3) - 4 = 18 - 4 = 14$ .

**Answer:**  $f(3) = 14$

---

7. Given  $f(x) = 1x + 3$ , find  $f(4)$ .

$$f(x) = 1x + 3, \quad x = 4$$

→ Substitute  $x = 4$ :  $f(4) = 1(4) + 3 = 4 + 3 = 7$ .

**Answer:**  $f(4) = 7$

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8. Given  $g(x) = 6x - 4$ , find  $g(-1)$ .

$$f(x) = 6x - 4, \quad x = -1$$

→ Substitute  $x = -1$ :  $g(-1) = 6(-1) - 4 = -6 - 4 = -10$ .

**Answer:**  $f(-1) = -10$

---

9. Given  $f(x) = 3x + -6$ , find  $f(-2)$ .

$$f(x) = 3x - 6, \quad x = -2$$

→ Substitute  $x = -2$ :  $f(-2) = 3(-2) + -6 = -6 + -6 = -12$ .

**Answer:**  $f(-2) = -12$

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10. Given  $g(x) = 5x - 8$ , find  $g(0)$ .

$$f(x) = 5x - 8, \quad x = 0$$

→ Substitute  $x = 0$ :  $g(0) = 5(0) - 8 = 0 - 8 = -8$ .

**Answer:**  $f(0) = -8$

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11. Given  $f(x) = 2x + -5$ , find  $f(1)$ .

$$f(x) = 2x - 5, \quad x = 1$$

→ Substitute  $x = 1$ :  $f(1) = 2(1) + -5 = 2 + -5 = -3$ .

**Answer:**  $f(1) = -3$

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12. Given  $g(x) = 7x - 8$ , find  $g(4)$ .

$$f(x) = 7x - 8, \quad x = 4$$

→ Substitute  $x = 4$ :  $g(4) = 7(4) - 8 = 28 - 8 = 20$ .

**Answer:**  $f(4) = 20$

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## Solving linear inequalities

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13. Solve the inequality:  $4x + 7 < 8$ . Express the answer as  $x < k$ .

$$4x + 7 < 8$$

→ Subtract 7 from both sides:  $4x < 8 - 7 = 1$ .

→ Divide both sides by 4:  $x < 1 / 4 = 1/4$ .

**Answer:**  $x < 1/4$

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14. Solve the inequality:  $3x - 7 \geq 1$ . Express the answer as  $x \geq k$ .

$$3x - 7 \geq 1$$

→ Add 7 to both sides:  $3x \geq 1 + 7 = 8$ .

→ Divide both sides by 3:  $x \geq 8 / 3 = 8/3$ .

**Answer:**  $x \geq 8/3$

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15. Solve the inequality:  $1x + 2 < 4$ . Express the answer as  $x < k$ .

$$1x + 2 < 4$$

→ Subtract 2 from both sides:  $1x < 4 - 2 = 2$ .

→ Divide both sides by 1:  $x < 2 / 1 = 2$ .

**Answer:**  $x < 2$

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16. Solve the inequality:  $4x - 5 \geq -1$ . Express the answer as  $x \geq k$ .

$$4x - 5 \geq -1$$

→ Add 5 to both sides:  $4x \geq -1 + 5 = 4$ .

→ Divide both sides by 4:  $x \geq 4 / 4 = 1$ .

**Answer:**  $x \geq 1$

---

17. Solve the inequality:  $2x + 5 < 8$ . Express the answer as  $x < k$ .

$$2x + 5 < 8$$

→ Subtract 5 from both sides:  $2x < 8 - 5 = 3$ .

→ Divide both sides by 2:  $x < 3 / 2 = 3/2$ .

**Answer:**  $x < 3/2$

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18. Solve the inequality:  $1x - 4 \geq -4$ . Express the answer as  $x \geq k$ .

$$1x - 4 \geq -4$$

→ Add 4 to both sides:  $1x \geq -4 + 4 = 0$ .

→ Divide both sides by 1:  $x \geq 0 / 1 = 0$ .

**Answer:**  $x \geq 0$

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19. Solve the inequality:  $2x + -7 < 5$ . Express the answer as  $x < k$ .

$$2x + -7 < 5$$

→ Subtract -7 from both sides:  $2x < 5 - -7 = 12$ .

→ Divide both sides by 2:  $x < 12 / 2 = 6$ .

**Answer:**  $x < 6$

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20. Solve the inequality:  $3x - 5 \geq 3$ . Express the answer as  $x \geq k$ .

$$3x - 5 \geq 3$$

→ Add 5 to both sides:  $3x \geq 3 + 5 = 8$ .

→ Divide both sides by 3:  $x \geq 8 / 3 = 8/3$ .

**Answer:**  $x \geq 8/3$

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21. Solve the inequality:  $4x + -6 < 3$ . Express the answer as  $x < k$ .

$$4x + -6 < 3$$

→ Subtract -6 from both sides:  $4x < 3 - -6 = 9$ .

→ Divide both sides by 4:  $x < 9 / 4 = 9/4$ .

**Answer:**  $x < 9/4$

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22. Solve the inequality:  $4x - 5 \geq -2$ . Express the answer as  $x \geq k$ .

$$4x - 5 \geq -2$$

→ Add 5 to both sides:  $4x \geq -2 + 5 = 3$ .

→ Divide both sides by 4:  $x \geq 3 / 4 = 3/4$ .

**Answer:**  $x \geq 3/4$

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23. Solve the inequality:  $6x + -8 < -6$ . Express the answer as  $x < k$ .

$$6x + -8 < -6$$

→ Subtract -8 from both sides:  $6x < -6 - -8 = 2$ .

→ Divide both sides by 6:  $x < 2 / 6 = 1/3$ .

**Answer:**  $x < 1/3$

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24. Solve the inequality:  $1x - 5 \geq 5$ . Express the answer as  $x \geq k$ .

$$1x - 5 \geq 5$$

→ Add 5 to both sides:  $1x \geq 5 + 5 = 10$ .

→ Divide both sides by 1:  $x \geq 10 / 1 = 10$ .

**Answer:**  $x \geq 10$

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## Slope between two points

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25. Find the slope of the line passing through (-4, 4) and (-3, 3).

$(-4, 4), (-3, 3)$

→ Use the slope formula:  $m = (y_2 - y_1) / (x_2 - x_1) = -1 / 1 = -1$ .

**Answer:**  $m = \frac{\Delta y}{\Delta x} = \frac{-1}{1} = -1$

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26. Find the slope of the line passing through (-2, 8) and (1, 7).

$(-2, 8), (1, 7)$

→ Use the slope formula:  $m = (y_2 - y_1) / (x_2 - x_1) = -1 / 3 = -1/3$ .

**Answer:**  $m = \frac{\Delta y}{\Delta x} = \frac{-1}{3} = -1/3$

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27. Find the slope of the line passing through (3, -8) and (11, 0).

$(3, -8), (11, 0)$

→ Use the slope formula:  $m = (y_2 - y_1) / (x_2 - x_1) = 8 / 8 = 1$ .

**Answer:**  $m = \frac{\Delta y}{\Delta x} = \frac{8}{8} = 1$

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28. Find the slope of the line passing through (2, -3) and (6, -1).

$(2, -3), (6, -1)$

→ Use the slope formula:  $m = (y_2 - y_1) / (x_2 - x_1) = 2 / 4 = 1/2$ .

**Answer:**  $m = \frac{\Delta y}{\Delta x} = \frac{2}{4} = 1/2$

---

29. Find the slope of the line passing through (-4, 1) and (3, 0).

$(-4, 1), (3, 0)$

→ Use the slope formula:  $m = (y_2 - y_1) / (x_2 - x_1) = -1 / 7 = -1/7$ .

**Answer:**  $m = \frac{\Delta y}{\Delta x} = \frac{-1}{7} = -1/7$

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30. Find the slope of the line passing through (0, 8) and (1, 8).

$(0, 8), (1, 8)$

→ Use the slope formula:  $m = (y_2 - y_1) / (x_2 - x_1) = 0 / 1 = 0$ .

**Answer:**  $m = \frac{\Delta y}{\Delta x} = \frac{0}{1} = 0$

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